Northeastern University, Khoury College of Computer Science

CS 6120 Natural Language Processing
Final Project Template

Due: November, 2025(100 points)

Caroline Zhang, Jinsha Lu, Nida Esen, Zijie Liu

PROJECT GIT REPOSITORY

1 Executive Summary and Abstract

This project presents MedRAG, an end-to-end medical question answering system built on Retrieval-Augmented Generation (RAG). The system retrieves relevant medical evidence from MedQuAD — a curated collection of NIH-aligned question-answer pairs — and grounds every generated answer in traceable source citations, directly reducing the risk of hallucination. We evaluate three retrieval strategies: sparse keyword search (BM25), dense semantic search using FAISS and sentence-transformers, and a hybrid approach combining both. Dense retrieval achieves the best Recall@5 of 0.747, outperforming BM25 (0.587) and hybrid (0.718). The final system is deployed as a Streamlit web application backed by a local Llama 3 LLM via Ollama, running entirely on GCP with no external API dependencies.

2 Motivation and Impact

2.1 RAG LLM Projects

Medical question answering demands accuracy and transparency. Large language models are well documented to hallucinate — generating plausible-sounding yet unsupported claims, which can cause genuine harm in clinical contexts.
MedRAG addresses this by grounding every answer in retrieved NIH-sourced passages with explicit citations, allowing users to independently verify outputs. This work provides a reproducible baseline for comparing BM25, dense, and hybrid retrieval on medical text.

3 Background and Related Work

Retrieval-Augmented Generation (RAG) (Lewis et al., 2020) combines a parametric LLM with a non-parametric retrieval component, grounding outputs in external evidence. It has

become the standard approach for knowledge-intensive NLP tasks where factual accuracy is critical.
We compare three retrieval strategies. **BM25** (Robertson & Zaragoza, 2009) is a keyword-based baseline that is fast but sensitive to vocabulary mismatch. **Dense retrieval** uses sentence-transformer embeddings with FAISS (Johnson et al., 2019) for semantic similarity search. **Hybrid retrieval** uses BM25 for first-stage recall and dense embeddings for reranking.

Dataset. We use **MedQuAD** (Ben Abacha & Demner-Fushman, 2019), a collection of over 47,000 NIH-sourced medical QA pairs available on Kaggle. Answer passages average ∼320 tokens while questions average ∼10 tokens, creating a vocabulary mismatch that motivates semantic retrieval.

4 Modeling Methodology

The MedRAG pipeline has four stages: (1) cleaning and chunking raw MedQuAD answers, (2) indexing via BM25 and/or FAISS, (3) retrieving the top-k chunks for a given query, and (4) generating a cited answer using Llama 3 via Ollama.
The default chunk size is 200 tokens with no overlap, which preserves medical terminology across passage boundaries. The LLM prompt instructs the model to answer only from retrieved context, cite every claim by `chunk_id` and `document_id`, and respond "I don't know" when no relevant passage is found — providing a direct hallucination-detection mechanism.
The project is fully containerized via Docker Compose, supporting both CPU and GPU environments. Running `docker compose up --build` builds the index and launches the Streamlit app on port 8501. The pre-built chunk artifact is committed to the repository for easy replication. Full instructions are at:
https://github.com/Ginsalulu-Jinsha/NIH-Medical-Retrieval-Augmented-Generation

5 Data and Data Analysis

5.1 Data Source

We use **MedQuAD** (Ben Abacha & Demner-Fushman, 2019), available on Kaggle. All NIH source attribution is preserved in the `source_page_url` column of every chunk.

5.2 Data Analysis and Exploration

The corpus contains over 47,000 QA pairs chunked into over 10,000 passages. Questions average ∼10 tokens; answers average ∼320 tokens — this asymmetry is the core retrieval challenge. Topic coverage spans diseases, symptoms, treatments, and genetics across 12 NIH domains.
Larger chunks with no overlap consistently outperform smaller or overlapping configurations under BM25, as small chunks fragment medical terms and overlap increases index noise.

6 Results and Evaluation

We evaluate using **Recall@5** and **Precision@5**. Recall@5 is the primary metric since complete answer coverage matters most for generation quality.

Table 6.1: Retrieval method comparison at $k = 5$

Method	Recall@5	Precision@5
BM25	0.5871	0.1375
Dense	**0.7469**	**0.1816**
Hybrid	0.7177	0.1746

Retrieval method comparison ($k = 5$). Dense retrieval achieves the best Recall@5 of 0.747. BM25 struggles due to vocabulary mismatch. Hybrid underperforms Dense because the BM25 candidate pool of 20 can exclude the correct document before reranking.

Table 6.2: Effect of chunk size on BM25 Recall@5 (overlap = 0 throughout)

Chunk Size	Overlap	Recall@5
60	0	0.5803
100	0	0.6020
150	0	0.6113
200	0	**0.6188**

Chunking analysis (BM25, $k = 5$). Retrieval strategy matters more than chunk size — the best BM25 configuration (0.619) still falls well below dense retrieval (0.747).

Qualitative example. For the query *"What are the symptoms of iron deficiency anemia?"*, the dense retriever surfaces an NIH passage describing fatigue, pallor, and shortness of breath. The system cites `chunk_id` and `document_id` for verification. For out-of-scope queries, the system responds: "I don't have sufficient information in the provided sources to answer this question."

7 Conclusions

MedRAG shows that RAG effectively reduces LLM hallucination in medical QA. Dense retrieval outperforms BM25 (0.747 vs. 0.587 Recall@5) and hybrid retrieval. Chunking hyperparameters matter less than retrieval strategy. Citation-grounded prompting provides a practical verifiability mechanism without additional tooling.
Future directions include increasing the hybrid candidate pool size, using biomedical encoders such as PubMedBERT, and surfacing retrieval confidence scores alongside generated answers.

8 Submission Guidelines

- **GCP Endpoint**: http://34.106.184.20:8501/
- **GitHub**: https://github.com/Ginsalulu-Jinsha/NIH-Medical-Retrieval-Augmented-
- **Team Members**: zhang.car@northeastern.edu, lu.jins@northeastern.edu, esen.n@northeastern.edu, liu.z4@northeastern.edu